Mubashir Ehsan

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EDUCATION

BS Computer Science — FAST NUCES Lahore *2025 – 2029*

– Courses: Object-Oriented Programming, Digital Logic Design, Programming Fundamentals.

FSc Pre-Engineering — KIPS Education System *2023 – 2025*

– College Topper & Position Holder — Academic Excellence Award.

PROJECTS

MediCore — Hospital Management System (C++, SFML) 2026

– Developed a 7,000+ line C++ hospital management system featuring Patient, Doctor, and Admin modules.
– Built a custom string library with 12+ functions (no STL strings), generic Storage template, and custom exception classes.
– Integrated SFML 3.0.2, file handling layer for persistence along security logging with login lockout after 3 failed attempts.

Smart Traffic Light Control System (Digital Logic, Logic Works 5) 2026

– Designed a 4-junction traffic control system using 74-series IC equivalents in Logic Works 5 (no microcontroller).
– Implemented a 2-bit Moore FSM using D flip-flops for signal sequencing with emergency vehicle priority via SR latch.
– Verified design using truth tables, K-maps, and timing diagrams across 9 test cases.

Company Management System (C++) 2025

– Developed an OOP-based system to manage departments and employee records.
– Implemented aggregation and composition and modular class-based design for scalability.

VOLUNTEERING & LEADERSHIP

Programming Competition Officer — SOFTEC 2026 *Feb 2026 – Present*
FAST NUCES Lahore

- Coordinated problem sets, logistics, and real-time execution for the programming competition track at one of Pakistan's largest student-run tech events.
- Managed cross-functional teams of officers, judges, and participants under high-pressure competitive conditions.

TECHNICAL SKILLS

• **Languages:** C++, SQL.

• **Core Concepts:** Object-Oriented Programming (OOP), Templates, Exception Handling, File Handling.

• **Digital Logic:** Boolean Algebra, FSM Design, Sequential Circuits, K-Maps, 74-Series ICs.

• **Tools & Technologies:** Git, GitHub, Visual Studio, CMake, SFML 3.0.2, Logic Works 5.